

Cybersecurity

Advanced Security Monitoring Tools

Lesson 19.2



Learning Objectives

By the end of today's class, you will be able to:

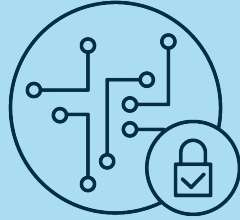
1 Differentiate between various advanced security monitoring solutions, such as SOARs, UBAs, and UEBA's, and determine which is most appropriate for a specific security situation.

2 Understand how knowledge of SIEM software and Splunk is valued in the information security job market.

3 Continue learning about Splunk with free Splunk training courses.



Welcome



Advanced Security Monitoring Tools

Introduction to Advanced Security Monitoring Tools

Today, we'll learn about other information security products used by security organizations.

We'll introduce Splunk's SIEM product, Enterprise Security, and explore several other security products available in the marketplace.

First

Next

We'll focus on security careers relevant to the knowledge and tools learned in the last five classes.

We will introduce and work towards the Splunk Fundamentals certification.

After the break

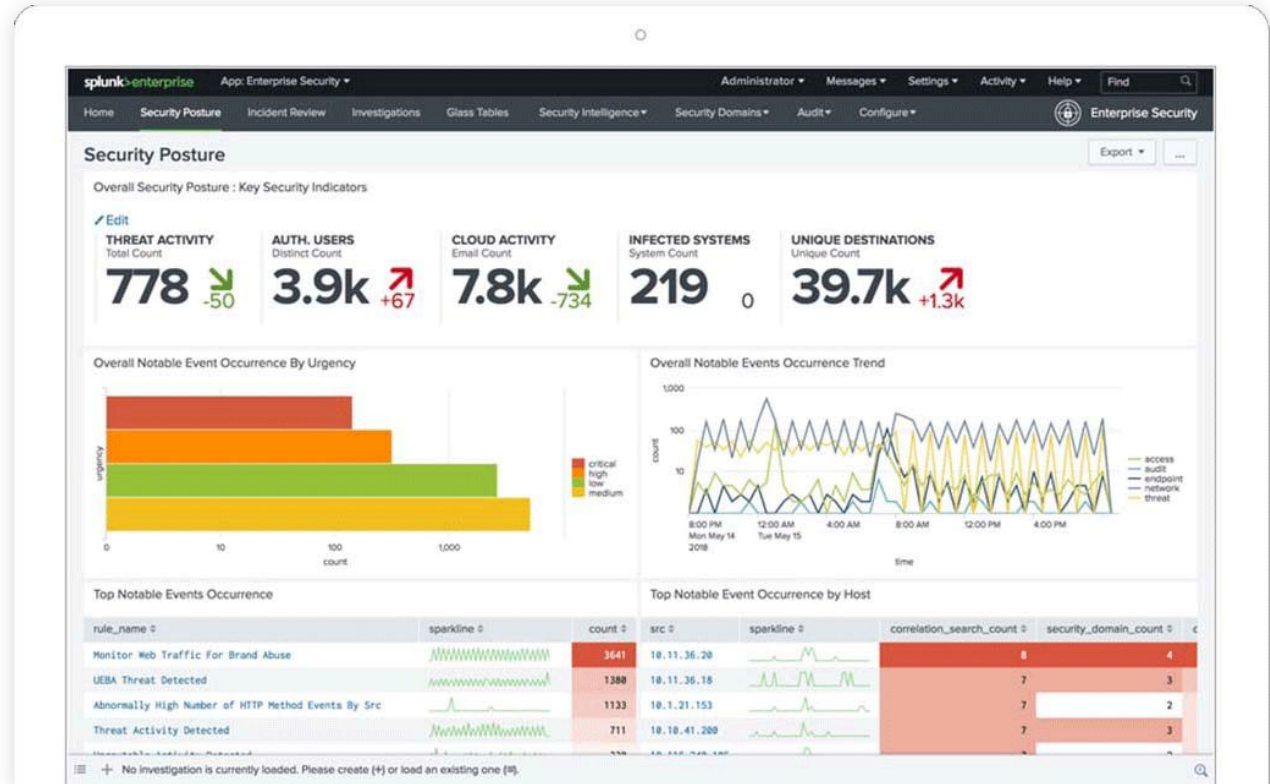
We'll introduce the final project, the bootCon presentation.

Finally

Splunk Enterprise Security

In the past five classes, we covered many of Splunk's capabilities and add-on applications.

The Splunk SIEM product, **Splunk Enterprise Security (ES)**, is one of the most popular add-on products for security professionals.

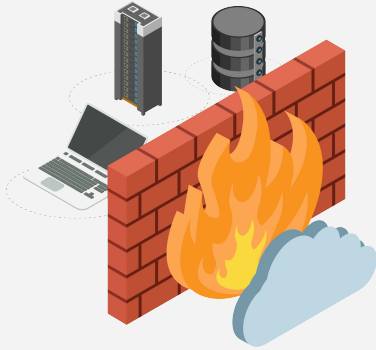


Splunk ES

Splunk ES is a SIEM product that provides security professionals with insights from machine-generated data, such as:

01

Network devices
like routers and firewalls



02

Endpoint devices
like antivirus solutions



03

**Vulnerability
management systems**
like Nessus



Splunk ES is one of the most popular add-on products for security professionals, as it has pre-built dashboards, reports, and built-in features.



Splunk ES

Splunk ES features allow you to:

01

Identify, prioritize, and investigate security events.

02

Gain insights into security events.

03

Monitor the status of a security environment.

04

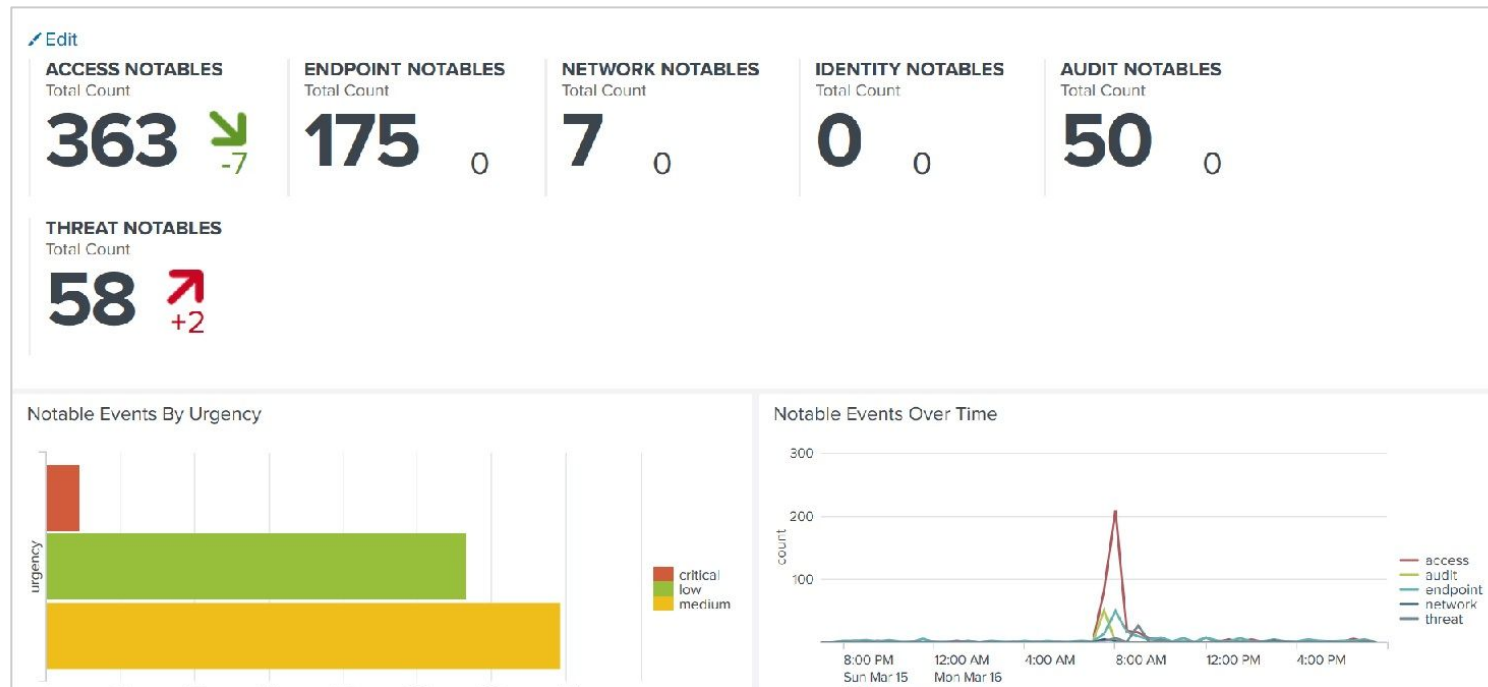
Audit security events.

05

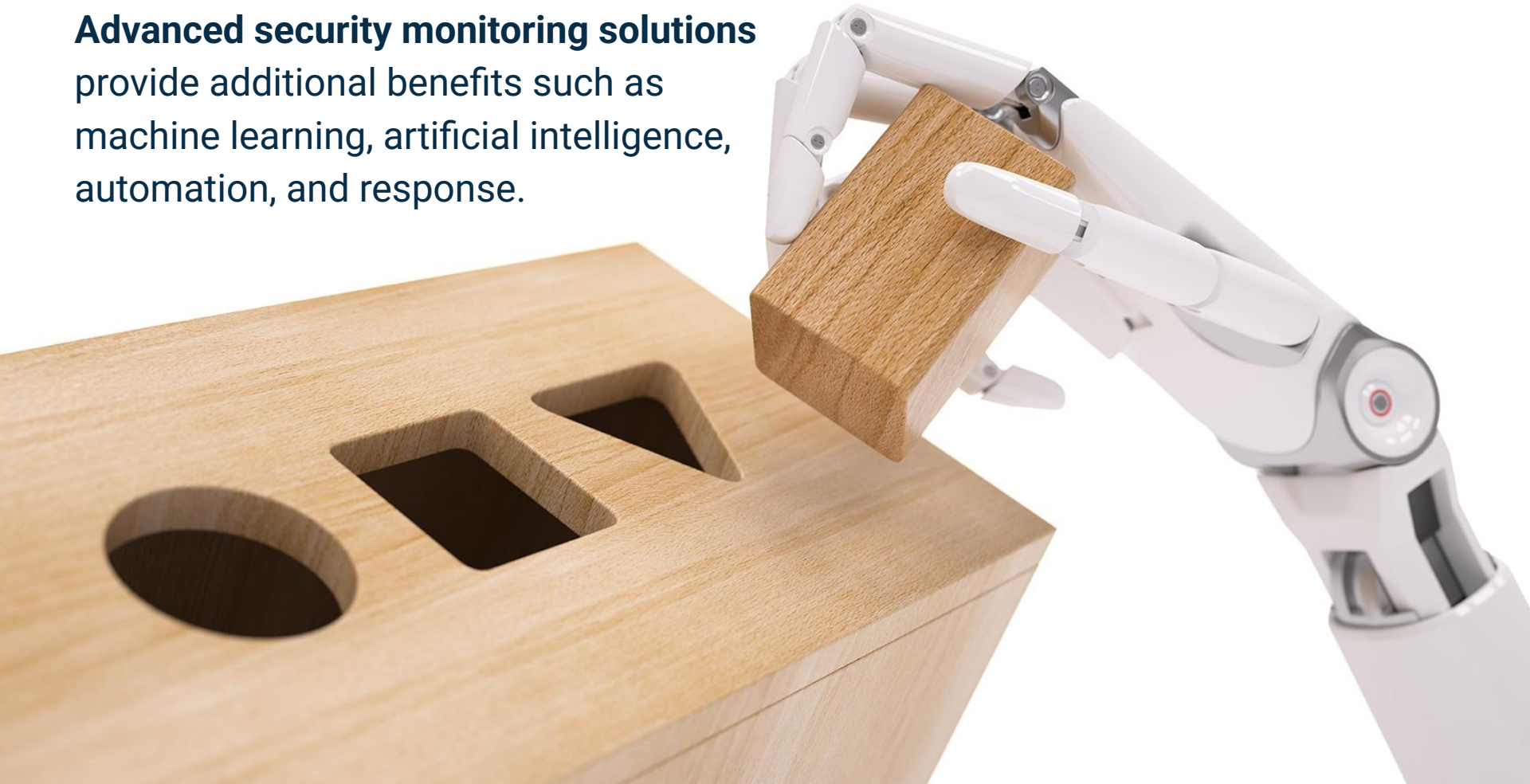
Navigate these tasks with a pre-built, easy-to-use interface.

Splunk ES

Example of a basic Splunk ES dashboard:



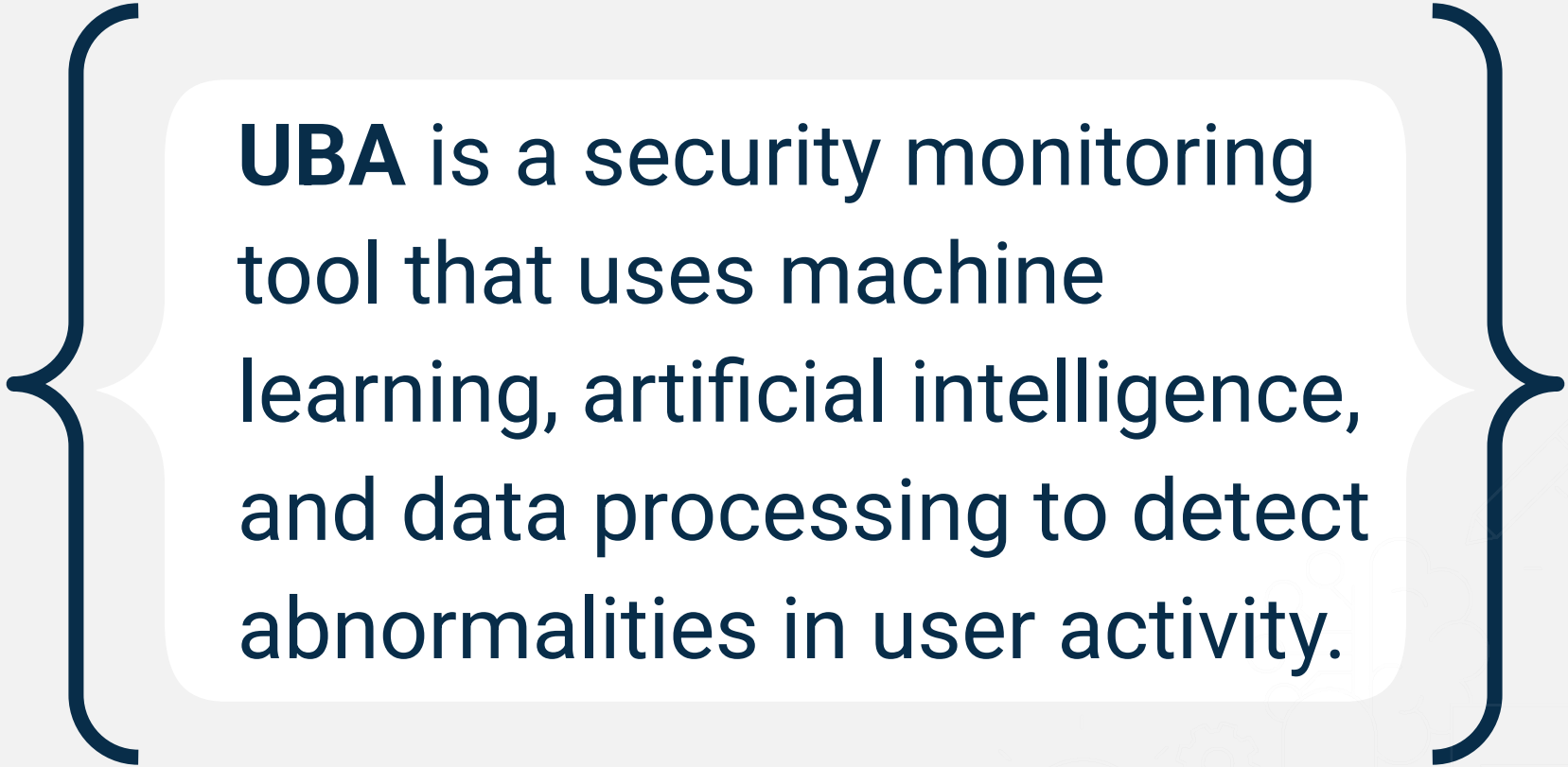
Advanced security monitoring solutions provide additional benefits such as machine learning, artificial intelligence, automation, and response.



Advanced Security Monitoring

The most popular advancements in the information security industry include:

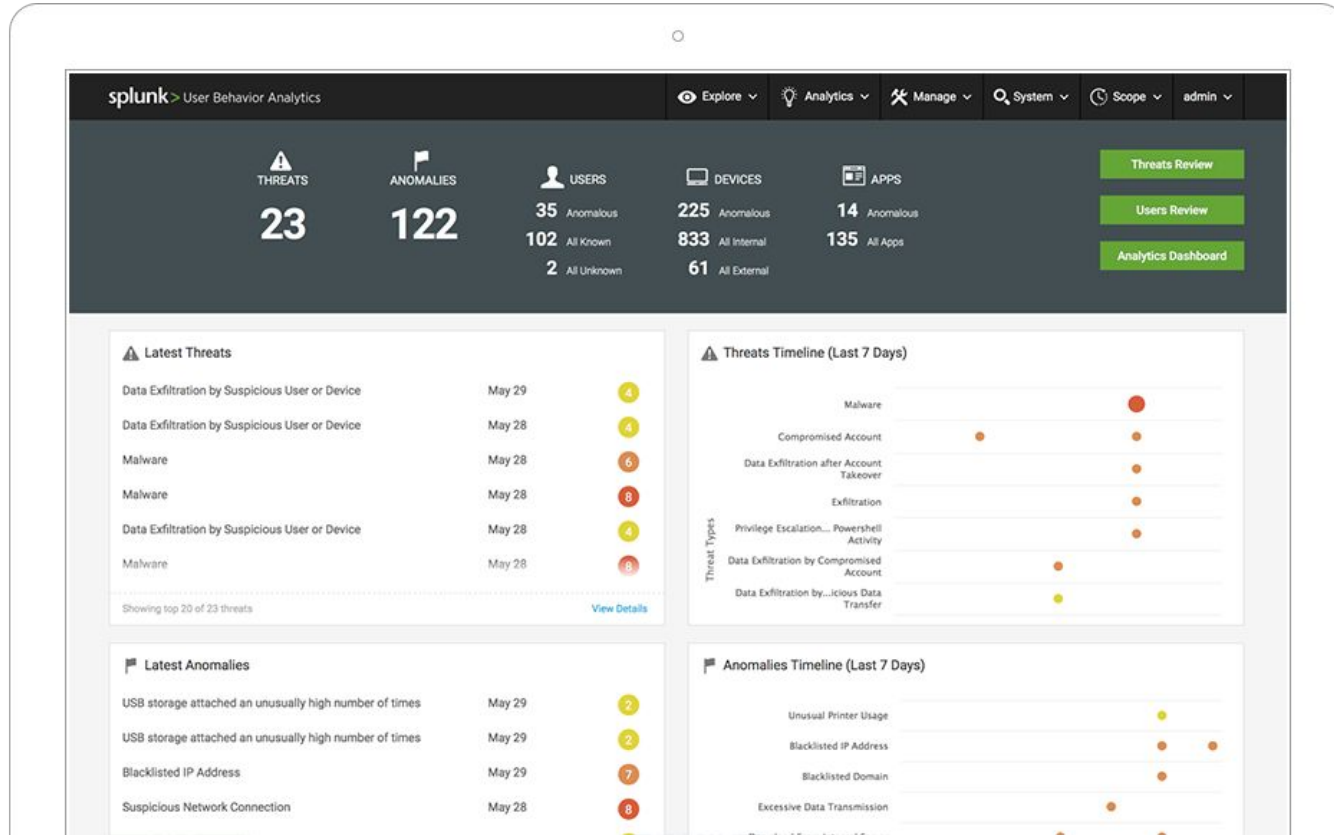
UBA	User behavior analytics
UEBA	User and entity behavior analytics
SOAR	Security orchestration, automation, and response
XDR	Extended detection and response



UBA is a security monitoring tool that uses machine learning, artificial intelligence, and data processing to detect abnormalities in user activity.

UBA

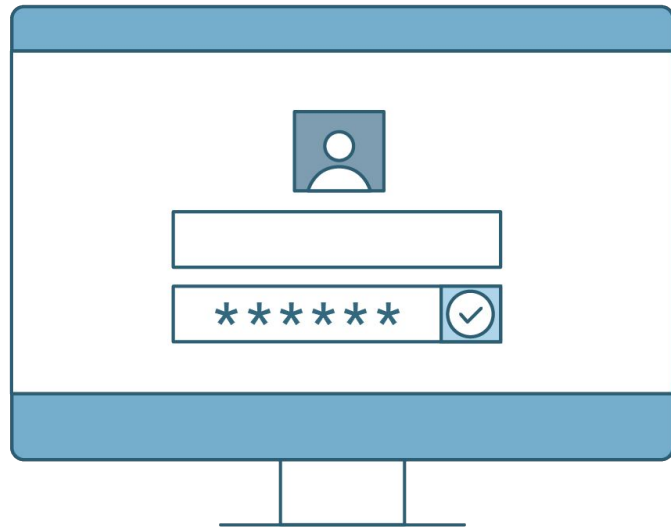
UBA gathers information about typical user behaviors and creates baselines.



UBA

An example of UBA in action:

1. UBA gathers information on the servers and systems that a user accesses, as well as when and how frequently they do so.
2. It create alerts for when a user's activity deviates from typical behavior
3. If a user usually only logs onto a server between 9am and 5pm Monday through Friday, UBA would create an alert if the user logged in on at 2am on a Saturday.





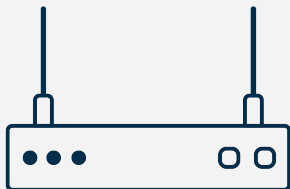
UEBA is a security monitoring tool similar to UBA, except it extends its monitoring to other "entities."

UEBA

Entities can include:

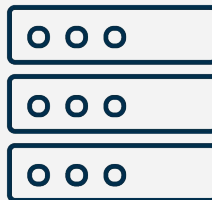
01

Routers



02

Servers



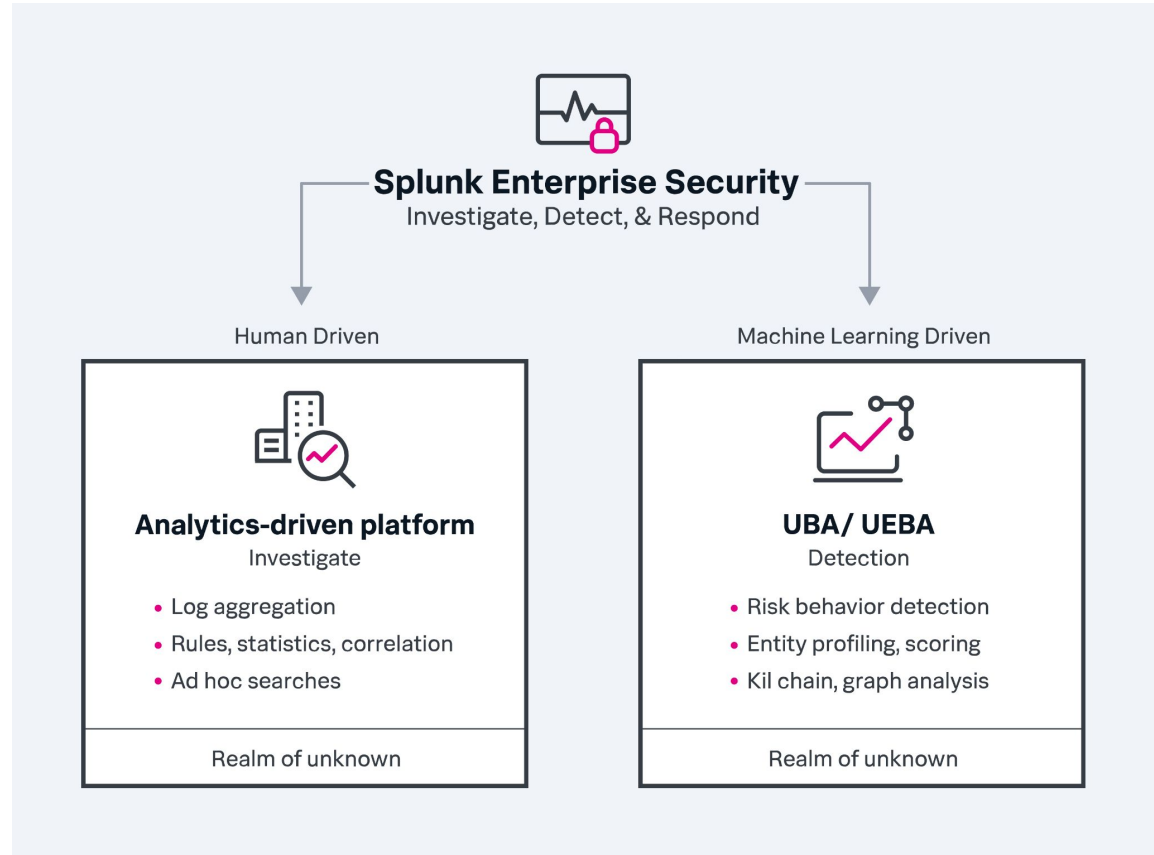
03

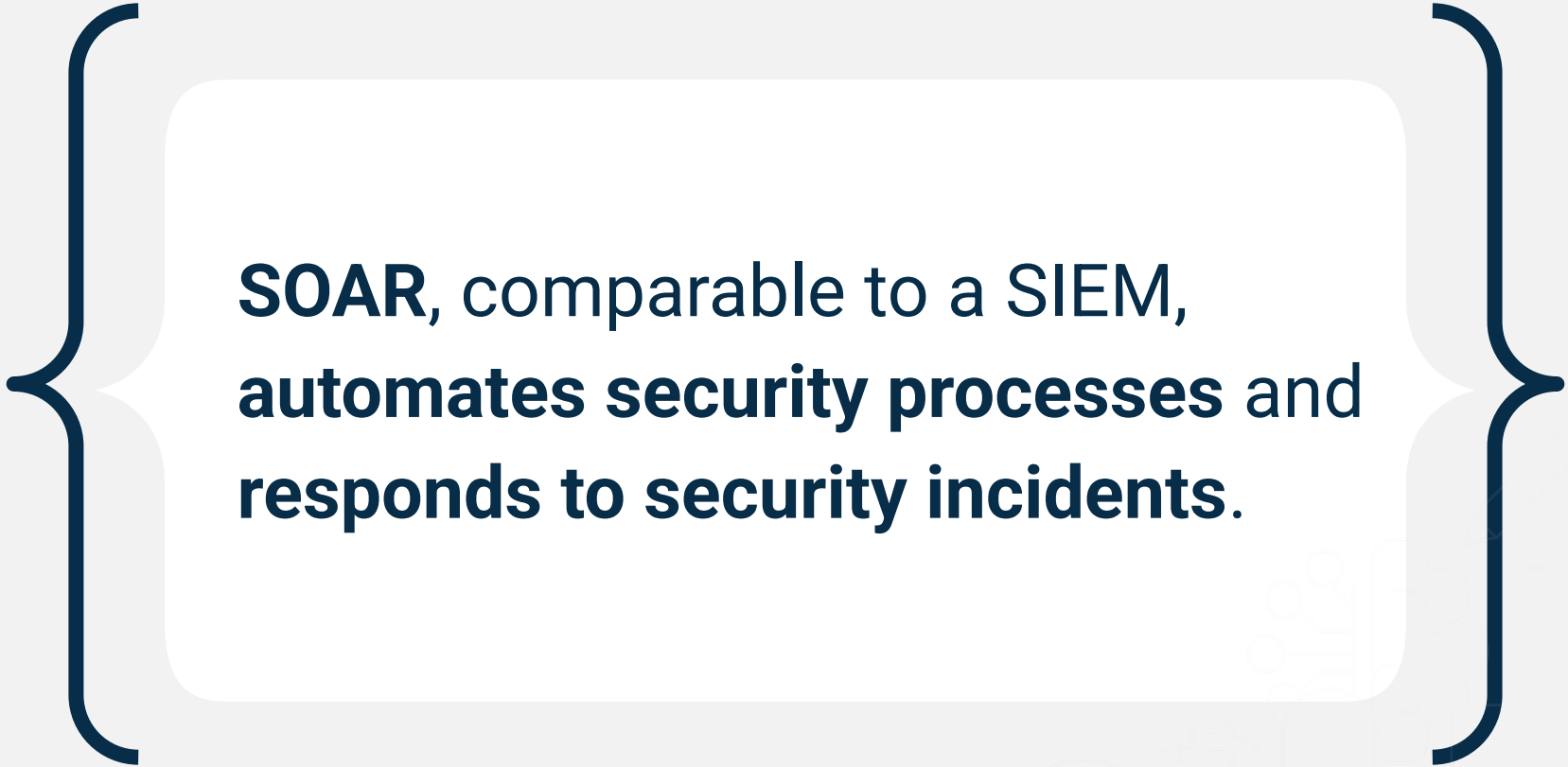
IoT Devices



UEBA

UEBA looks at typical user and entity behaviors and creates alerts when they display unusual activity.



A graphic featuring a white rounded rectangle with a dark blue border, flanked by large dark blue curly braces on both sides. The text inside the rectangle is in a dark blue, sans-serif font. The background is light gray with faint, stylized icons of a person, a gear, and a lightbulb.

SOAR, comparable to a SIEM,
automates security processes and
responds to security incidents.

SOAR

Automating security processes examples:

Creating logging

Assigning priorities to
security incidents

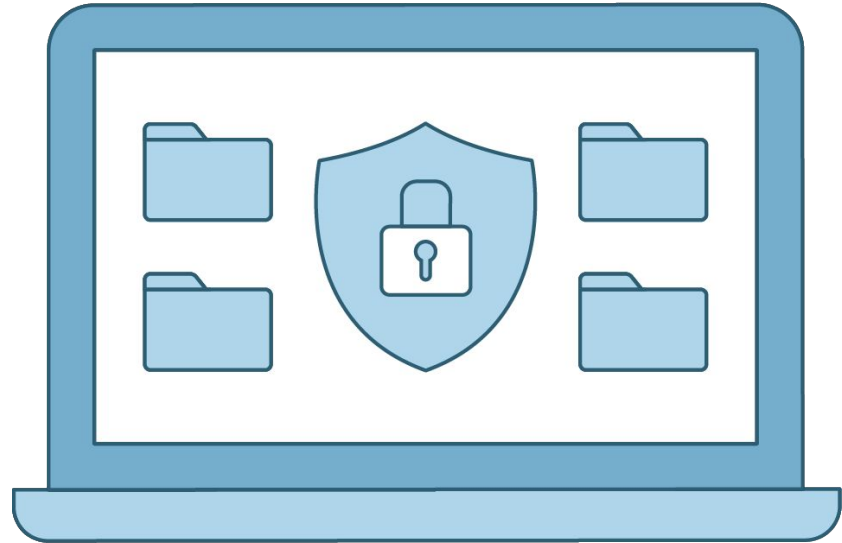


SOAR

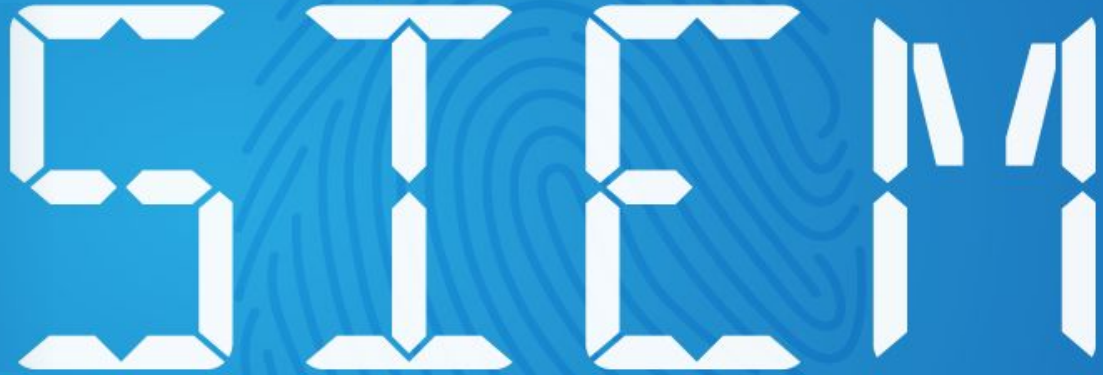
Responding to security incidents examples:

Launching security investigations

Mitigating threats



Similar to a SIEM, SOAR gathers machine data from multiple entities and analyzes the data for security events.

The letters 'SIEM' are rendered in a large, white, stylized font with a 3D effect. The background features a faint, circular fingerprint pattern in a lighter shade of blue, centered behind the text.

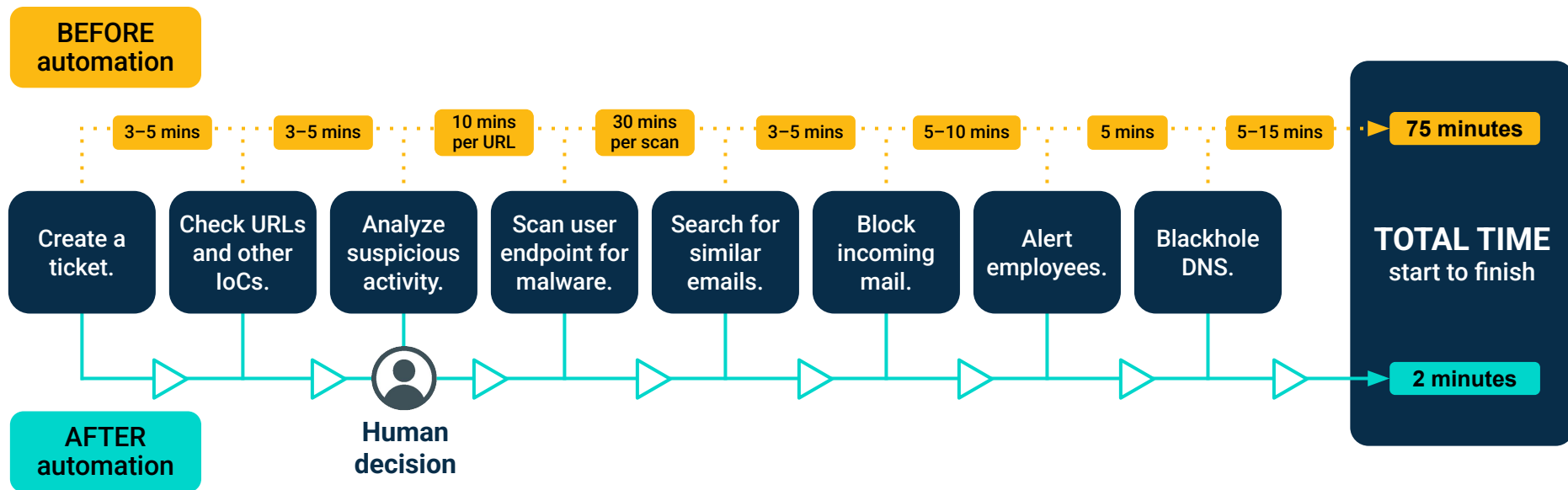
SIEM



SOAR uses **playbooks** that detail the processes and response actions for specific events.

SOAR

This diagram illustrates how using SOAR playbooks can decrease incident response time. Playbooks are uniquely designed and configured by each organization.



XDR

Extended detection and response

- XDR offers advanced detection, rapid response, and intuitive automation.
- It provides improved cross-domain threat hunting and forensic investigation capabilities.
- It lacks the log management, retention, and compliance capabilities of SIEM.





Activity:

Advanced Security Monitoring Tools

OMP management has approved a larger budget for the SOC to enhance its security monitoring.

- You are tasked with researching several SOAR, UBA, and UEBA vendors to provide a summary to your management about the products you are recommending.

Suggested Time:
20 Minutes



Time's up!
Let's review



Questions?





Splunk Careers

Now, we'll explore careers and certifications related to the Splunk knowledge and tools learned over the past five lessons.



Splunk experience is a
desirable qualification for
many information security
positions.



Splunk Careers: InfoSec Positions

SOC analysts

SOC analysts work in a security operations department alongside security engineers. Their positions involve detecting, containing, and potentially remediating information security threats. Most SOC analysts use SIEM products, such as Splunk ES, to monitor their environment.

Cyber threat analysts

Cyber threat analysts analyze an organization's networks and applications to protect organizations from cybercriminals. Cyber threat analysts often use Splunk products to make predictions about cybercriminals and their potential attacks.

Application security engineers

Application security engineers Use Splunk to fix web and mobile application vulnerabilities. AppSec engineers use Splunk to analyze their application logs to assist with creating and testing their remediation.

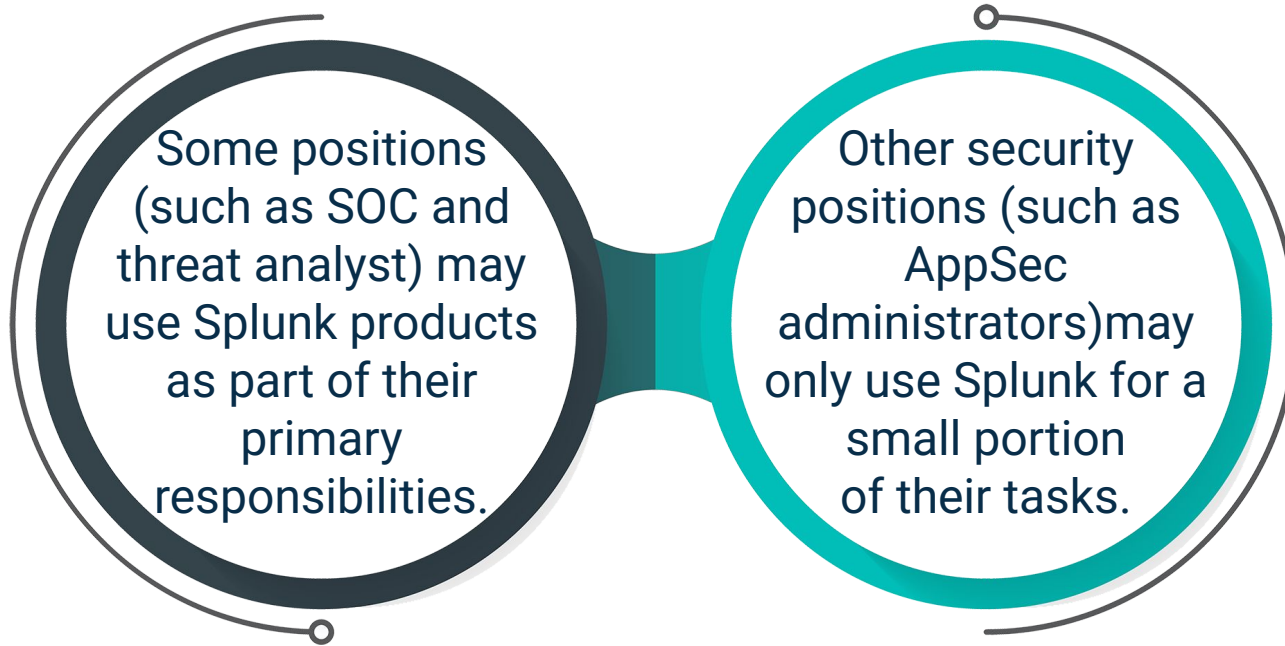
Network security administrators

Network security administrators use products like Splunk to monitor suspicious network traffic, such as DDoS attacks. They can use the findings from Splunk logs to mitigate and prevent future attacks.

Incident response managers

Incident response managers use Splunk to monitor the status of ongoing security investigations when an incident has occurred.

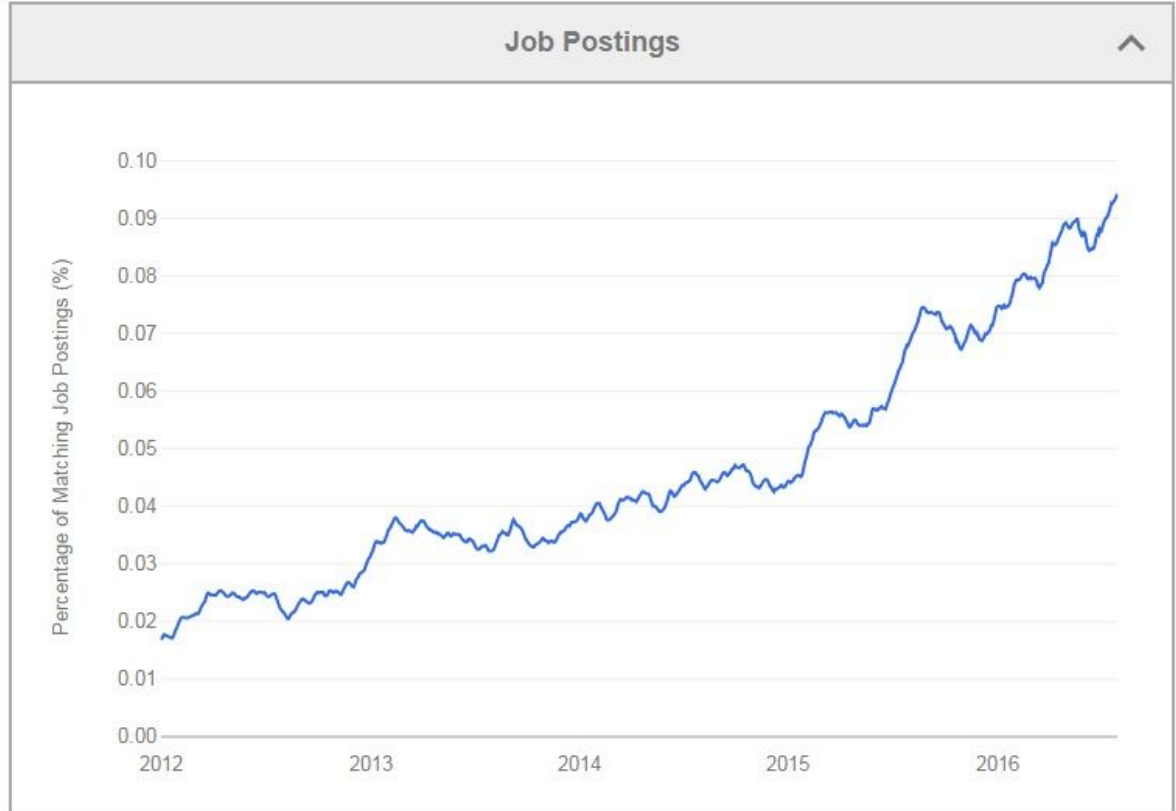
Splunk Careers: InfoSec Positions



Either way, understanding how to use Splunk is a valuable skill for InfoSec professionals.

Splunk in InfoSec Careers

Splunk is already a required skill in many roles, and the industry demand is increasing every year.





Activity:

Splunk Careers

In this activity, you will search several job sites for Splunk-related careers and answer questions about each position.

Suggested Time:
15 Minutes





Time's up!
Let's review



Questions?





Break
15 mins



SIEM Certifications

Splunk Certifications

Having a certification can help a cyber professional acquire a new position, receive a promotion, and attain networking opportunities with professionals who have similar certifications.





Splunk offers several certifications of varying skills.

Splunk Certifications

Splunk offers many certifications for a variety of skill levels:



Splunk Core Certified User

Entry-level certification that demonstrates a user's basic ability to use the Splunk software



Splunk Core Certified Power User

Demonstrates a user's foundational skills with Splunk's core software, plus more complex skills, such as creating calculated fields and data models



Splunk Core Certified Advanced Power User

Demonstrates a user's capability to design reports, complicated searches, and dashboards

Splunk Certifications

Splunk offers many certifications, for a variety of skill levels.



Splunk Enterprise Certified Admin

Focuses on an individual's ability to support daily administrative tasks using Splunk ES



Splunk Enterprise Certified Architect

Focuses on a Splunk administrator's role supporting advanced troubleshooting, configurations, and deployments in Splunk ES



Splunk Enterprise Security Certified Admin

Focuses on a Splunk administrator's role to support installation, advanced troubleshooting, configurations, and deployments in Splunk ES

Splunk Certifications

Like for many certifications in the InfoSec field, training for Splunk certifications is expensive.

Fortunately, Splunk offers many single-subject courses for free.

The screenshot displays the Splunk Training & Certification website. The browser address bar shows the URL: splunk.com/en_us/training.html?sort=Newest&filters=filterGroup1FreeCourses. The website header includes the Splunk logo and navigation links: Products, Solutions, Why Splunk?, Resources, Support, and a 'Free Splunk' button. The main banner features the text 'Splunk Training + Certification' and a sub-header: 'Hungry for bite-sized training? Check out our menu of new, single-subject courses!' with a 'Learn More' button. Below the banner, a 'QUICK LINKS' section provides links to 'My Training Profile', 'Splunk Education Student Handbook', 'Splunk Certification Handbook', 'FAQ', 'Authorized Learning Partners', 'Videos', and 'Basic Subscription Datasheet'. A search bar and a 'FILTERED BY: Free Courses' filter are present. The 'Filter all' section shows 29 results, with 'Free Courses' selected. The main content area displays three free courses: 'Splunk User Behavior Analytics', 'IT Essentials Learn - Walkthrough', and 'Free Splunk Fundamentals 1'. Each course card includes a brief description and a 'Learn More' link.

QUICK LINKS | [My Training Profile](#) | [Splunk Education Student Handbook](#) | [Splunk Certification Handbook](#) | [FAQ](#) | [Authorized Learning Partners](#) | [Videos](#) | [Basic Subscription Datasheet](#)

Search FILTERED BY: Free Courses x

Filter all
29 Results [Clear All](#)

Content Type ^

- ☐ Courses
- ☒ Free Courses
- ☐ Certification Exams

Certification v

Role v

Product v

Suite v

FREE COURSE

Splunk User Behavior Analytics

In this course we tour the UBA interface, look at how UBA defines threats, discuss steps to take when responding to possible threats and how to triage false positives.

[Learn More >](#)

COURSE

IT Essentials Learn - Walkthrough

This 8-minute eLearning course steps you through the IT Essentials Learn app.

[Learn More >](#)

FREE COURSE

Free Splunk Fundamentals 1

This course teaches you how to search and navigate in Splunk, use fields, get statistics from your data, create reports, dashboards, lookups, and alerts.

[Learn More >](#)



Activity:

Splunk Certification

In this activity, you'll register for a Splunk account and begin single-subject courses.

Suggested Time:
60 Minutes



Time's up!
Let's review

Activity: Splunk Certifications

Resources for taking the Core Certified User certification exam:

1

Splunk: [Core Certified User certification](#)

2

Splunk: [Journey to getting certified \(video\)](#)

3

Splunk: [Exam registration tutorial](#)

4

Pearson VUE: [Exam registration](#)

5

Splunk: [Certification study guide](#)



Splunk Certifications

If you are interested in continuing your education towards a **Splunk Core Certified User** certification:

You can access additional learning courses in your own time.



You can take the Splunk Core Certified User exam to earn a Splunk certificate.



Questions?





Overview of Remaining Projects And Introduction to **BootCon**

Project Weeks

You will complete two more projects in this course:

1. Defensive Security Project

- This project will occur at the end of the Defensive Security module and will incorporate everything you learnt in this module.
- We'll cover the details next week.

2. Final Project: BootCon

- This project will occur in the last week of class.
- We'll get an overview now so that you can begin brainstorming and preparing.

Cybersecurity Research Presentations

Cybersecurity professionals commonly present the following to their peers:

- 1 Security research they're conducting
- 2 Newly discovered security vulnerabilities of products, devices, software, or hardware
- 3 Demonstrations of the “hacks” that will exploit these vulnerabilities
- 4 Mitigations to protect against these vulnerabilities



- These presentations often occur at security conferences, trade shows, and other industry events.
- Cybersecurity professionals commonly attend these events to remain up to date with new technological developments and techniques.

BootCon Presentations

Similarly, you will have an opportunity to showcase your skills during bootCon.

1 On the last day of class, we will hold a cybersecurity conference called **bootCon**.

2 Each student will have an opportunity to showcase the skills they learned during the bootcamp with a presentation.

3 If you elect to present as a group, every group member must participate in the presentation.

4 Most cyber professionals present at conferences when they've found a new vulnerability. However, for bootCon presentations, it is acceptable and recommended that you recreate a finding that has already been discovered.



A bootCon presentation
is **NOT** a research paper.



While research
will be required,
all presentations
must be tangible
and demonstrable.

**A demonstration
can either be:**

- Conducted in person
- A prerecorded video
that accompanies
the presentation
*(if a live demonstration
isn't practical)*



BootCon Presentations

Each bootCon presentation should fall into one of the following three categories:

	01	02	03
Category	Exploiting a vulnerability of an IOT device	Developing code or a program that can complete a cybersecurity task	Demonstrating how a cybersecurity tool that was not covered in class can accomplish a specific goal
Example	Hacking your personal Blu-Ray player	Developing a Python script that can automate an Nmap scan	Using SET to design a social engineering campaign



Questions?





BootCon Rules and Requirements

BootCon Rules and Requirements

You must submit a project summary to your instructor for approval.
The summary should be submitted with a Slack message by Week 22.

Your project summary should include:

- 1 The topic and title of your presentation
- 2 The end goal or vulnerability being exploited
- 3 A list of devices and/or technologies that will be used to accomplish the goal
- 4 A summary of how the devices and technologies will be used to accomplish the goal

BootCon Rules and Requirements



Under no circumstances may any aspect of your bootCon presentation be unethical or illegal.

You must:

- 1 Perform all hacks and tests in simulated environments.
- 2 Complete any network connections in your home and/or controlled environment.
- 3 Only perform IoT hacks on devices that you own.

BootCon Rules and Requirements

Presentations must have a goal whose achievement you can demonstrate.

For example:

Goal:

Cracking WEP wireless traffic
from your home router



Demonstration:

Showing how you captured and
cracked your wireless traffic



You can either conduct your demonstration live or record it and present it while you guide people through what took place.

BootCon Rules and Requirements

You must submit your presentation in the form of a Google Slides deck that, at a minimum, includes the following:

Cover slide	Presentation title and team member(s) presenting
Technical background	• Explanation of why you selected the topic you are presenting • Networking, cryptographic, or security concepts applied • Research steps taken
Demonstration preview	Preview of the steps that you'll take in the upcoming demonstration
Demonstration	A live or recorded demonstration
Demonstration summary	A summary of the demonstration and any impact that it may have
Mitigation	Recommendations for mitigating against the attack that you just conducted (not required if your presentation isn't about an attack)

BootCon Rules and Requirements

01

You will present your project on the last day of class, and your total presentation time should be 7–10 minutes.

02

If you choose to form a group of 2–3 students, presentation time can be 5 minutes per student.

BootCon Rules and Requirements

If you need ideas for your presentation, refer to the [bootCon presentation guide](#), which contains:

- 1 Sample presentations
- 2 List of Kali Linux tools
- 3 List of IoT hacks
- 4 Videos of hacks presented at security conferences



Remember: Your project proposals are due in Week 22!



Questions?





Next class

Designing Your Defensive Solution



Thank you